Vincent Zhen

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Education

University of Waterloo — Waterloo, ON

Candidate for Bachelor of Computing and Financial Management — *September 2024 – April 2029*

- Recipient of the University of Waterloo President's Scholarship of Distinction
- Achieved Excellent Academic Standing with Term Distinction for a **92% CGPA**
- Awarded the **KPMG Scholarship** for outstanding and exemplary academic achievement

Experience

Software Developer — Toronto, ON

Pepper (Pomegranate Technologies, Inc.) — *April 2025 – August 2025*

- Expanded Pepper's Smart Attribute Library Taxonomy platform (SALT) to centralize, enrich, and standardize product data for foodservice distributors, encompassing **3M+** unique items
- Scraped **300,000+** catalogue items from major foodservice distributors using Playwright, extracting GTINs, images, allergens, and nutrient data
- Automated AI-powered website creation with Python, n8n, Snowflake, and 10Web, securing **$10,000+ ARR** in distributor contracts and advancing **10+ active leads**
- Spearheaded a large-scale image reordering project using OpenAI Vision to reorganize **1M+ images** across **400,000+** unique items in SALT, improving thumbnail image accuracy to over **95%**
- Built an in-house AI image generation pipeline with AWS S3, OpenAI, Gemini, and MongoDB, added a fuzzy matching step using OpenAI embeddings and scikit-learn to align requests with **SALT's commodity items**, reducing redundant image generation and boosting catalogue engagement
- Designed and deployed **ETL** pipelines for Syndigo, Frosty Acres, and Essendant via API/SFTP integrations, ingesting **450,000+** new items into MongoDB and AWS S3 using scheduled AWS Glue runtimes

STEM Workshop Lead — Markham, ON

STEMed Initiative — *May 2023 - June 2024*

- Co-founded an organization aimed at teaching STEM concepts with in-person interactive workshops
- Utilized Arduino and Micro:Bit kits to deliver hands-on STEM learning with real-time constructive feedback
- Hosted **4** comprehensive coding and experiment-based workshops, engaging and educating over **100** students

Projects

Robo-Advisor | *Python, NumPy, pandas, yfinance, Matplotlib*

- Designed a technical analysis algorithm to generate the most profitable portfolio from a list of given tickers
- Led a team of 2 in developing ticker filters, volatility indicators, and correlation indicators for the algorithm
- Analyzed **2+ years** of historical stock data with Pandas and NumPy to uncover correlations and visualize those insights using Matplotlib

Patient Intake Form | *Typescript, Next.js, Tailwind CSS, Supabase, GraphQL*

- Developed a modern, production-ready patient intake platform using Next.js, GraphQL, and Supabase to streamline healthcare data management.
- Built validated multi-field forms and secure file uploads with React-Hook-Form, Zod, and Vercel Blob.
- Created an admin dashboard with SSR and AG Grid for fast, filterable patient record management.
- Designed optimized PostgreSQL schemas and type-safe GraphQL APIs for reliable, maintainable performance.

Frequency Filter to Combat Hearing Loss | *Python, Typescript, Matplotlib, Next.js*

- Built a modern website using Next.js and TypeScript, featuring seamless audio file upload integration
- Crafted a simple yet practical GUI with Tkinter, visualizing frequency fluctuations with Matplotlib
- Received the **Best Beginner Hack award** for IgnitionHacks out of **43** teams

Technical Skills

Languages: Python, C, C++, SQL, HTML/CSS, JavaScript, Racket, TypeScript, Bash
Tools: AWS S3, AWS Glue, MongoDB, Supabase, OpenAI, Gemini, Git, Next.js, Excel, Holistics BI, n8n, Make
Libraries: pandas, requests, Scikit-learn, boto3, NumPy, Matplotlib, Playwright